



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

GAZIPUR-1704, DHAKA, BANGLADESH

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

PROJECT REPORT

EEE 4518

ELECTRICAL AND ELECTRONIC WORKSHOP

***PROJECT TITLE:* Electrically Controlled PCR Machine**

SUBMITTED BY:

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1. Introduction

The Polymerase Chain Reaction (PCR) machine, also known as a thermocycler, is a critical laboratory instrument widely used in the biomedical and molecular biology fields. It is designed to precisely control the temperature cycles required for the amplification of DNA samples. By repeatedly varying the temperature through specific stages—denaturation, annealing, and extension—the PCR machine enables a small segment of DNA to replicate exponentially, producing millions of copies within a short period.

PCR technology is one of the most fundamental techniques in modern molecular biology. It plays an essential role in medical diagnostics, cancer research, genetic analysis, forensic science, biotechnology, and infectious disease detection. Because the DNA amplification process is highly temperature-sensitive, accurate and stable thermal control is the most important functional requirement of a PCR machine. Any deviation in temperature can significantly affect the efficiency and reliability of DNA amplification.

While commercial PCR machines are highly effective, they are often expensive and structurally complex. As a result, designing and constructing a simplified PCR system serves as an excellent exercise in applied engineering and embedded system design. It allows the integration of temperature sensing, control systems, power electronics, and user interface components into a single functional device.

In this project, a prototype PCR machine is designed and implemented using a microcontroller-based control system. The system focuses on developing a reliable temperature regulation mechanism through a controlled heating system and thermal feedback loop. A thermal block is used to hold the PCR tubes while ensuring uniform heat distribution, and temperature sensors are employed to continuously monitor the thermal conditions. The control unit processes this information and regulates the heating mechanism accordingly to achieve the desired thermal cycling process.

Additionally, the system incorporates a display interface to present real-time temperature and operational status to the user, while a Switch Mode Power Supply (SMPS) provides efficient and stable power for the entire setup. Through this design, the project demonstrates the practical implementation of temperature control techniques and system integration required for building a functional PCR thermocycler prototype.

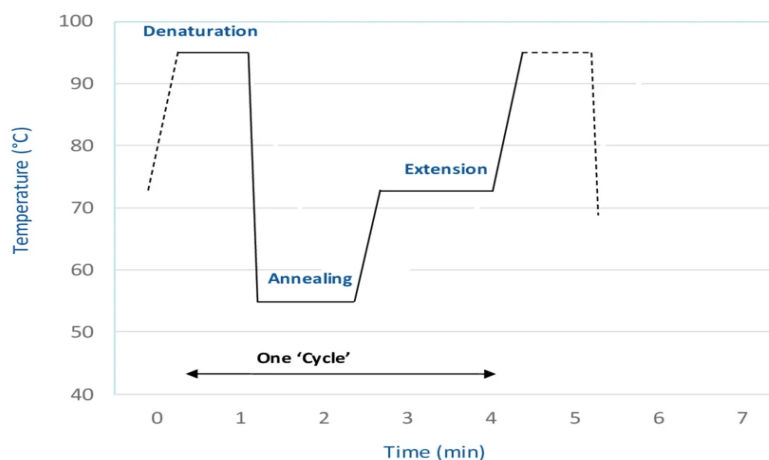


Figure 1: Temperature profile of a single PCR amplification cycle.

2. Objectives

The main objectives of this project are:

- To understand the working principle and operational stages of a PCR machine.
- Segment different parts of PCR machine for implementation
- To design and develop a thermal block capable of holding PCR tubes and ensuring uniform heat distribution.
- To implement a heating mechanism capable of achieving the required temperature ranges for PCR operation.
- To develop a temperature sensing and feedback system for accurate thermal monitoring.
- To integrate a microcontroller-based control system using a relay module for temperature regulation.
- To implement an LCD display interface to show system temperature and operational status.
- To implement an LCD display interface to show system temperature and operational status.

3. Required Components

The following hardware components were used in the design and implementation of the PCR machine prototype:

- **Aluminum Thermal Block:** Custom-fitted to hold the PCR tubes, ensuring uniform heat distribution across all samples during thermal cycling.
- **Switch Mode Power Supply (SMPS), 12V DC 10A:** Acts as the primary power source, delivering the high current necessary to drive the heating elements and the cooling system.
- **Rechargeable 9V Battery:** Provides a dedicated, stable power source for the microcontroller, isolating the sensitive logic circuitry from the high-current thermal components.
- **High-Power Resistors (10W, 10 Ω):** Serve as the primary resistive heating elements to rapidly raise the temperature of the thermal block during the denaturation and extension phases.
- **Thermoelectric Cooler (TEC / Peltier Module):** Utilizes the Peltier effect to provide active, rapid cooling to the thermal block, which is essential for quickly dropping the temperature for the annealing stage.
- **Aluminum Heatsink:** Attached to the hot side of the Peltier module to absorb and transfer waste heat away from the system.
- **BLDC Fan (12V):** Mounted to the heatsink to force active airflow, dissipating the heat and maintaining the cooling efficiency of the Peltier module.
- **Arduino Uno:** Serves as the central control unit. It reads real-time sensor data and executes the logic required to automate the heating and cooling cycles.
- **MAX6675 Temperature Sensor with K-Type Thermocouple:** Continuously measures the exact temperature of the aluminum block and provides precise digital feedback to the microcontroller.
- **2-Channel Relay Module:** Acts as an electronic switch, allowing the low-voltage Arduino to safely turn the high-current heating resistors and Peltier module on and off.

- **LCD Display (with Adapter):** Provides a real-time user interface, displaying the current block temperature, cycle phase, and overall system status.
- **DC-DC Buck Converter:** Steps down the main 12V supply to lower, regulated voltages safely required by specific electronic peripherals.
- **Essential Accessories:** Includes standard assembly and testing tools such as a soldering station, hot glue gun, thermal paste (crucial for efficient heat transfer between the block and thermal components), jumper wires, a digital multimeter, a screwdriver set, and a laptop for programming the Arduino.

4. Circuit Diagram

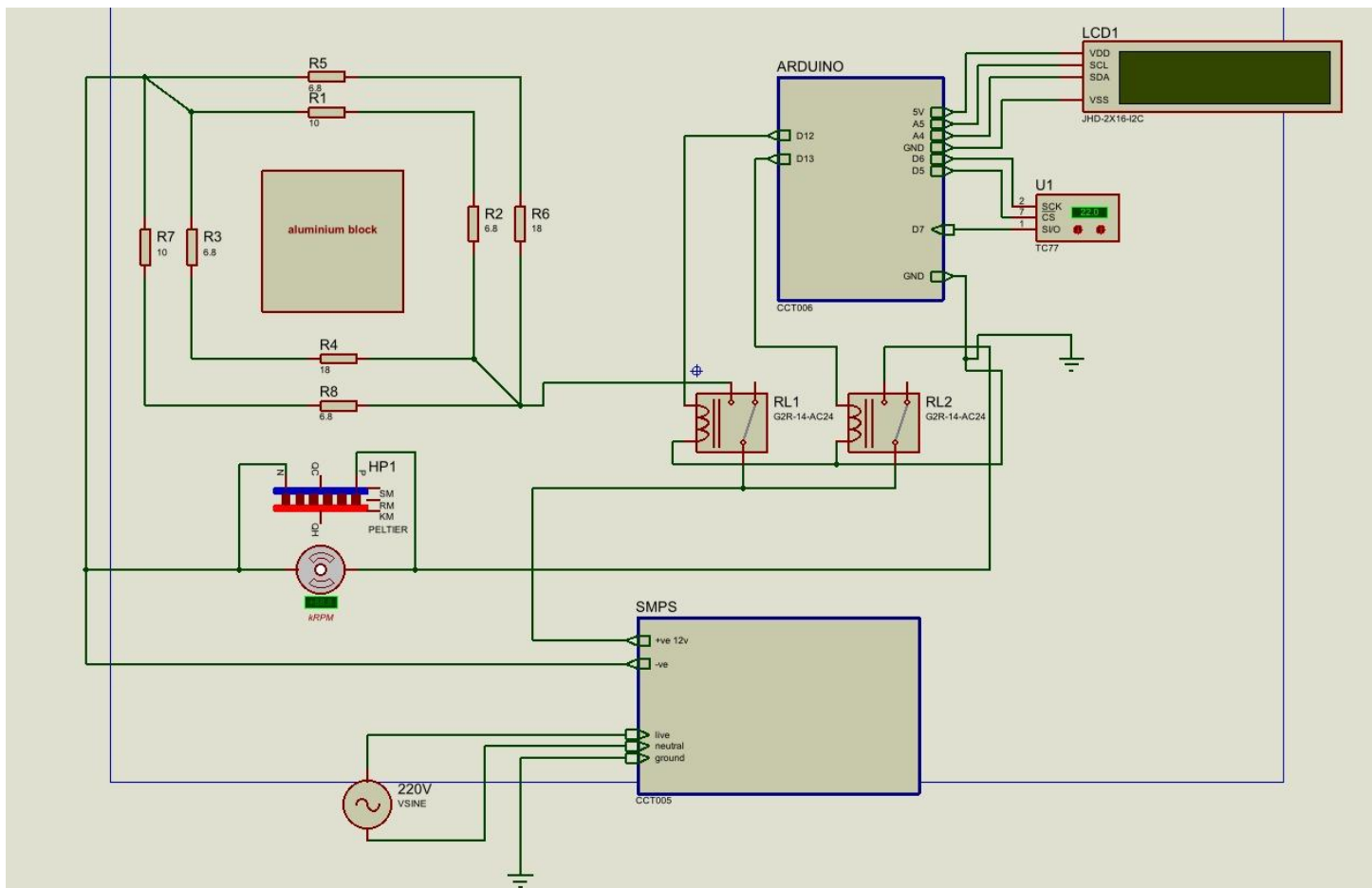


Figure 2: Circuit Diagram of the whole system

4.1 Explanation

- **Power Distribution:** The system is powered by a standard 220V AC mains supply, which is stepped down and rectified by a Switch Mode Power Supply (SMPS). The SMPS provides a highly stable 12V DC output (rated up to 10A) to handle the heavy load of the heating and cooling elements. To safely power the control logic, a DC-DC buck converter steps the 12V supply down to 9V for the Arduino

Uno. Furthermore, a rechargeable 9V battery is wired in parallel to the microcontroller. This acts as a localized Uninterruptible Power Supply (UPS), ensuring the Arduino remains active and retains its thermal cycling memory during sudden power outages or load shedding.

- **Thermal Actuation (Heating and Cooling):** The thermal manipulation of the aluminum block is handled by two distinct circuits, both governed by the microcontroller via a 2-channel relay module:
 - **Heating Circuit:** An array of high-power resistors (R1 through R8) is arranged around the aluminum block. When heating is required (during the denaturation phase), the Arduino triggers the first relay (RL1), allowing the 12V supply to pass through the resistor network, rapidly generating heat.
 - **Cooling Circuit:** For the active cooling phase (annealing), a Peltier module and an associated 12V DC cooling fan are wired in parallel. When a temperature drop is required, the Arduino activates the second relay (RL2). This simultaneously turns on the Peltier module to draw heat away from the block and the fan to exhaust that waste heat.
- **Control Logic and User Interface:** The Arduino Uno functions as the central controller of the system. It continuously receives temperature data from the MAX6675 temperature sensor module connected to a K-type thermocouple placed near the aluminum thermal block.

Based on the measured temperature and predefined PCR cycle parameters, the Arduino controls the relay module to switch the heating and cooling circuits on or off, forming a closed-loop temperature control system.

For user interaction, an I2C-based LCD display is connected to the Arduino. The display provides real-time information such as the current temperature and operational status of the PCR cycle, enabling easy monitoring of the system.

5. Features

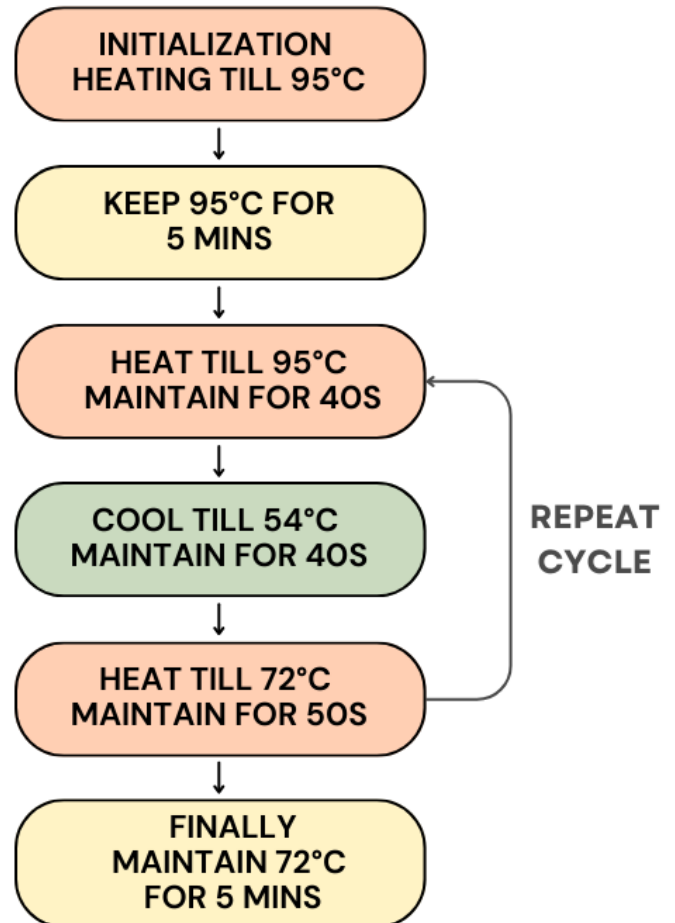
The main features of the developed PCR machine prototype are listed below:

- **Moderate Temperature Control** – The system can maintain required temperature levels for different PCR stages through controlled heating and cooling.
- **Adequate Heating and Cooling Mechanism** – The combination of high-power resistors for heating and a Peltier-based cooling system enables effective thermal cycling.
- **Uniform Temperature Distribution** – The aluminum thermal block helps distribute heat evenly across the PCR tubes, improving temperature stability.
- **Real-Time Status Display** – An LCD display provides continuous monitoring of system temperature and operational status.
- **Programmable Temperature States** – The microcontroller allows predefined temperature stages to be implemented for the PCR thermal cycling process.

6. Working Principle

6.1. Flow Chart

PCR Machine Working Principle



6.2 Explanation

To achieve DNA amplification, the machine must accurately execute the following temperature-dependent stages:

- a. **Initialization:** At the beginning of the process, the reaction mixture is heated to a high temperature, typically between 94–98°C, for several minutes. This step ensures the complete separation of double-stranded DNA into single strands and activates the DNA polymerase enzyme.
- b. **Thermal Cycling:**

- **Denaturation (94–98°C):** The double-stranded DNA separates into two single strands.
- **Annealing (50–65°C):** Primers bind (anneal) to their complementary DNA sequences.
- **Extension (72°C):** DNA polymerase extends the primers to synthesize new DNA strands.

c. **Finalization:** After completion of all cycles, the reaction is maintained at 72°C for approximately 5 minutes to ensure complete extension of all amplified DNA fragments.

For our test cycle in the lab our parameter was-

- Initialization:** The system heated the thermal block to 95°C and maintained it for 5 minutes.
- Thermal Cycles:**
 - Denaturation:** The block was maintained at 95°C for 40 seconds
 - Annealing:** The Peltier module actively cooled the block to 54°C, holding it for 40 seconds.
 - Extension:** The heating elements raised the temperature to 72°C, maintaining it for 50 seconds
- Finalization:** The system held a steady 72°C for 5 minutes to conclude the automated process.

7. Code

The code that was used for controlling the heating cycle uploaded to Arduino UNO:

```
#include "max6675.h"
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27,16,2);
int s0 = 7;
int thermoCS = 5;
int thermoCLK = 6;
int c;

int i=1;
int j=0;
int k=0;

MAX6675 thermocouple(thermoCLK, thermoCS, s0);
```

```

unsigned long start, final, elapsed;

void setup() {
  Serial.begin(9600);

  //Serial.println("MAX6675 test");
  // wait for MAX chip to stabilize
  delay(500);
  lcd.init();
  lcd.backlight();

  pinMode(13, OUTPUT);
  pinMode(12, OUTPUT);
  digitalWrite(13,HIGH);
  digitalWrite(12,HIGH);

}

void loop() {

  if(i==1){
    //Serial.println("start");
    lcd.setCursor(0, 0);
    lcd.print("initial");
    temp(95,300);
    while(k<35){
      /*Serial.print("cycle no.");
      Serial.println(k);*/
      lcd.setCursor(0, 0);
      lcd.print("cycle no, " + String(k));
      temp(95,40);
      temp(54,40);
      temp(72,50);
    }
  }
}

```

```

        k=k+1;
    }
    //Serial.println("final");
    temp(72,300);
    i=0;
}

else if(i==0){
    /*Serial.print("C = ");
    Serial.print(thermocouple.readCelsius());
    Serial.print(" F = ");
    Serial.println(thermocouple.readFahrenheit());
    delay(300);*/
    digitalWrite(13,HIGH);
    digitalWrite(12,HIGH);
    lcd.setCursor(0, 0);
    lcd.print("C = ");
    lcd.print(String(thermocouple.readCelsius())+" ");
    lcd.print(" F = ");
    lcd.print(String(thermocouple.readFahrenheit()));
    delay(300);

    //!!!!!! For the MAX6675 to update, you must delay AT
    LEAST 250ms between reads
}
}

void temp(int th, int tt){
    int e=0;
    c=thermocouple.readCelsius();
    //Serial.print("C = ");
    //Serial.println(c);
    while (c<th){
        digitalWrite(13,HIGH);

```

```

digitalWrite(12, LOW);
delay(300);
c=thermocouple.readCelsius();
lcd.setCursor(0,1);
lcd.print(String(c)+"to"+String(th));
//Serial.println(c);
//Serial.println("heating");
}
while(c>th){
    digitalWrite(13, LOW);
    digitalWrite(12, HIGH);
    delay(300);
    c=thermocouple.readCelsius();
    lcd.setCursor(0,1);
    lcd.print(String(c)+"to"+String(th));
    //Serial.println(c);
    //Serial.println("cooling");
}
if(c>th-1 && c<th+1){
    start=millis();
    /*Serial.println(c);
    Serial.println("maintaining");*/
    digitalWrite(13, HIGH);
    digitalWrite(12, HIGH);
    while(e<tt){

        if (c>th){
            digitalWrite(13, LOW);
            delay(1000);
            digitalWrite(13, HIGH);

        }
        else if(c<th){
            digitalWrite(12, LOW);

```

```

        delay(1000);
        digitalWrite(12,HIGH);

    }
    delay(300);
    c=thermocouple.readCelsius();
    //Serial.println(c);
    final=millis();
    elapsed=final-start;
    e=int(elapsed)/1000;
    //Serial.println(e);
    lcd.setCursor(0,1);
    lcd.print(String(c)+" time"+String(th)+"s");
}
}
delay(300);
}

```

8. Laboratory Testing and Results

During data collection and testing in the test with side by side with commercial [miniPCR](#) imported from USA.



a. Objective: To evaluate the performance of the prototype PCR/Thermal Cycler

b. Experimental Setup:

- Cycle number: 35 cycles
- Reaction volume: 25 μ l

- Expected amplicon size: 353 bp
- Temperature sensing method: Thermocouple embedded in aluminum block
- Control system: Arduino IDE
- Heating mechanism: Resistive Elements

c. Observation and Results

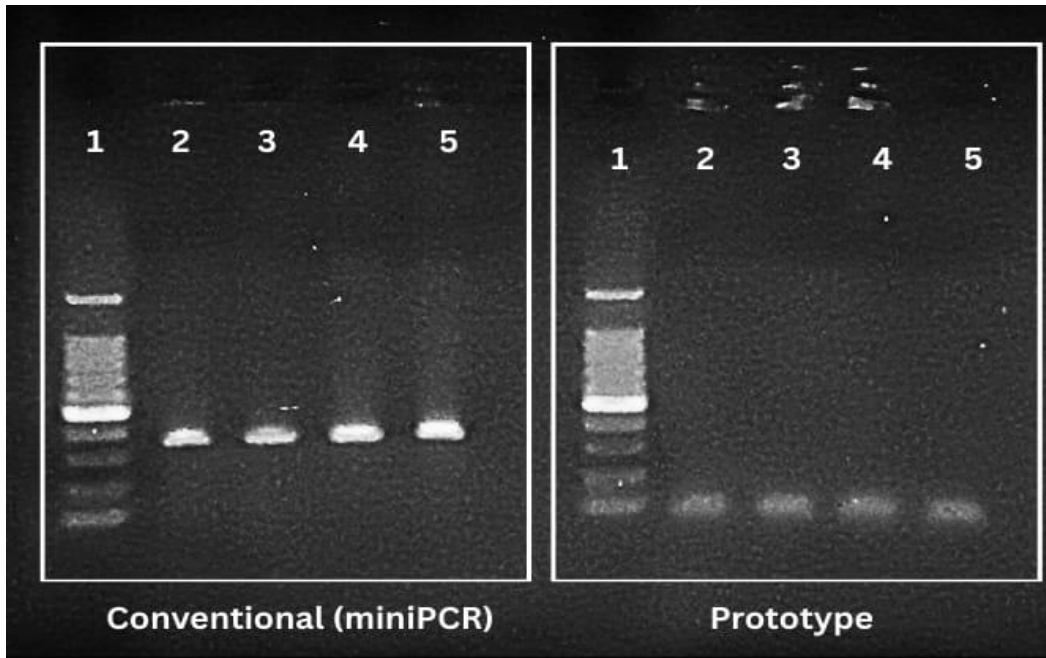


Figure: Comparative gel electrophoresis results. The left panel shows amplification from the Commercial system (miniPCR); the right panel displays amplification from the Prototype system. [Legend: Lane 1 contains a 100bp DNA ladder for size estimation. Lanes 2-5 contain targeted DNA samples from identified patients.]

d. Data Comparison:

Based on the gel electrophoresis results (Figure 1), the performance of the two systems was observed as follows:

Commercial Thermocycler (miniPCR):

- Expected 353 bp band clearly observed in all reaction lanes.
- Demonstrated clear and specific amplification of the target sequence.
- Exhibited high band intensity, indicating robust DNA synthesis.

Prototype PCR Machine:

- Amplification was successfully observed.
- The generated product size was approximately 100 bp.
- Demonstrated lower band intensity compared to the commercial system.
- No significant non-specific smearing was observed, indicating a systematic rather than random thermal event.

e. Technical Analysis

The discrepancy in amplicon size suggests a temperature calibration deviation between block temperature and actual reaction temperature inside the PCR tube.

Potential technical causes:

- Block-to-sample thermal gradient
- Heat transfer inefficiency through plastic PCR tubes
- Thermal inertia mismatch
- PID control tuning limitations
- Absence or inefficiency of heated lid

f. Performance Metrics

- Cycle completion: Successful
- Amplification capability: Functional
- Specificity: Reduced
- Thermal accuracy: Requires calibration
- Repeatability: Yes

g. Recommendations for System Optimization

- Perform in-tube thermocouple calibration.
- Implement temperature offset compensation.
- Optimize PID tuning parameters for more precise and stable temperature control.
- Improve thermal contact pressure between the tubes and the heating block.
- Integrate or refine heated lid mechanism.
- Validate the system's ramp rate and steady-state thermal stability.

9. Problem Faced:

- Initially, the entire heating system was designed and implemented using an AC supply. The resistive heating elements were directly powered using AC, and temperature control was attempted under this configuration. However, during practical implementation, significant temperature instability was observed.

To overcome this issue, the system was redesigned to operate fully on DC. By converting the power stage to a DC-based configuration, smoother and more stable power delivery was achieved. This modification significantly improved temperature stability and allowed more precise control of the heating process.

- Another major issue encountered during the project was inaccurate temperature data from the thermocouple sensor. The temperature readings were inconsistent and did not match expected or reference measurements.

To resolve this problem, a calibration procedure was performed. The thermocouple readings were compared against a known reference temperature source, and correction factors were applied where necessary. After calibration, the sensor output became stable and aligned with expected values, improving the accuracy and reliability of the temperature control system.

- During the early stage of hardware implementation, one of the Switch Mode Power Supply (SMPS) circuits was accidentally damaged. As this was our first practical experience working with an SMPS-based power system, certain design and handling aspects were not fully optimized during initial testing. After replacing the damaged SMPS, greater care was taken in circuit verification, connection checking, and staged power-up procedures to prevent similar failures. This experience significantly improved our understanding of power electronics safety and practical implementation considerations.

10. Conclusion

This project successfully demonstrated the design and construction of a functional, low-cost PCR thermocycler. By integrating a custom aluminum thermal block, microcontroller-based logic, and a bidirectional heating and cooling mechanism, the prototype achieved the precise temperature regulation strictly required for DNA amplification. The system effectively automated the critical denaturation, annealing, and extension cycles, proving the functional viability of both the hardware and software architecture. Ultimately, this build establishes a robust engineering foundation for accessible laboratory equipment. It validates that complex thermal cycling can be achieved using affordable, readily available components without sacrificing core functionality. With future refinements—such as advanced thermal insulation and optimized control calibration—this prototype has the strong potential to reach performance metrics comparable to high-end, commercial-grade thermocyclers.

11. References:

- <https://www.instructables.com/Arduino-PCR-thermal-cycler-for-under-85/>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10021165/>
- <https://www.labmanager.com/understanding-thermal-cyclers-an-introduction-and-overview-31635>

GitHub Repository Link:

<https://github.com/faiadsarthok/PCR-Machine-Prototype-Arduino-Thermocycler-.git>

12. Contributions

Name And Student ID	Contribution
Mubeen Raisul Hassan Chowdhury (220021308)	Design and construction of the aluminum thermal block; design and implementation of the power supply circuit; electrical soldering; biomedical lab liaison; final system testing.
Nahid Sadique Tasin (220021115)	Design of the hardware setup and control circuitry; development of the software logic and thermal algorithms; final system testing.
Hussaib Hazzil Abbasi (220021104)	Procurement of materials; physical construction of the hardware and control circuits; research and calculation of system parameters; final system testing.
M Faiad Faisal Sarthok (220021116)	Project documentation and presentation preparation; physical construction of the hardware and control circuits; system debugging and optimization; final system testing
An-Nafi Ahasan (220021152)	Design and implementation of the power supply circuit; physical construction of the hardware and control circuits; project management and budget oversight; final system testing.

13.CO PO Covered

- knowledge profile:
 - **K3:** As we need theoretical knowledge of power system and electronics to make the whole system and programming knowledge for software development.
 - **K5:** We are designing a system fully on a principle that follows thermodynamics and the systems requires electronics component to achieve the results.
 - **K7:** As our goal is giving the genomic researchers an economical alternative solution.
- Problem Solving profile:
 - **P1:** Fulfills K3, K5 & K7 of Knowledge Profile
 - **P3:** Our system can heavily depend on the scale as for small scale we won't even need high power supply but if consider high scale it may need continuous power supply and won't be available for standby mode. For that we may have to introduce a backup mode.
 - **P7:** We need various type of component which will interact with each other to complete the whole project
- Activities:
 - **A1:** We need many types of resources as per people with knowledge, then materials like aluminum block, Electronics, power supply, etc. Information like the feasible market value of the project and practical usage.
 - **A2:** Engineering fields of mechanical, electrical and Software skills are needed.
 - **A4:** As there are no current manufactures of this type of machine in Bangladesh

Course Outcomes					
COs	CO Statement	Corresponding POs	Assigned K P A	Covered K P A	Covered
CO1	Analyze and evaluate	PO2: Problem analysis PO4: Investigation PO7: Environment and sustainability	K3, K4 P3 A3	K3 P3	Yes
CO2	Design & develop	PO3: Design/development of solutions	K5, K6 P2 A3, A4	K5 A4	Yes
CO3	Use modern tools to find solutions to real world problems.	PO5: Modern tool usage	K4, K6 P2, P4, A1, A3	A1	Yes
CO4	Explain effectively (both in oral and written form) the design and implementation of the project.	PO9: Individual work and teamwork PO10: Communication PO11: Project management and finance PO12: Lifelong learning	K7, P1, P3, A1, A3, A4	K7 P1, P3, A1, A4	Yes